

Summary report

Wellbore: 15/9-F-1 C

Period: 2014-03-07 00:00 - 2014-03-08 00:00

Status:	normal
Report creation time:	2018-05-03 13:51
Report number:	15
Days Ahead/Behind (+/-):	2.3
Operator:	Statoil
Rig Name:	MÆRSK INSPIRER
Drilling contractor:	Maersk Drilling
Spud Date:	2014-02-21 14:30
Wellbore type:	
Elevation RKB-MSL (m):	54.9
Water depth MSL (m):	91
Tight well:	Y
HPHT:	Y
Temperature (I):	
Pressure (I):	
Date Well Complete:	2014-03-15

Dist Drilled (m):	0
Penetration rate (m/h):	-999.99
Hole Dia (in):	12.25
Pressure Test Type:	formation integrity test
Formation strength (g/cm3):	1.55
Dia Last Casing (I):	

Depth at Kick Off mMD:	
Depth at Kick Off mTVD:	
Depth mMd:	3056
Depth mTVD:	2873.1
Plug Back Depth mMD:	
Depth at formation strength mMD:	2517
Depth At Formation Strength mTVD:	2454.8
Depth At Last Casing mMD:	2507.7
Depth At Last Casing mTVD:	2445.8

Summary of activities (24 Hours)

RIH with 9 5/8" casing from 2337 m to 2410 m. Changed handling equipment to 10 3/4".
M/U X-over joint and RIH with 10 3/4" casing stands from 2410 m to 2902 m.
Installed casing hanger and RIH on HWDP landing string. Landed casing at 3047 m.
Circulated prior to cement job.
Performed cement job.

Summary of planned activities (24 Hours)

Pressure test casing.
WOC prior to bullhead B-annulus to brine.
Bullhead B-annulus, set and P/T 10 3/4" seal assembly.
POOH with PADPRT, LD same.
Repair annular BOP.

Operations

Start time	End time	End Depth mMD	Main - Sub Activity	State	Remark
00:00	00:45	2410	drilling -- casing	ok	RIH with 9 5/8" 53,5 ppf P110 VamTop casing from 2337 m to 2410 m. Filled casing every 5th joint. Average running rate 8 joints/hr.
00:45	01:00	2410	drilling -- casing	ok	Performed TBT for changing handling equipment.
01:00	03:15	2410	drilling -- casing	ok	R/D BX elevators. B/O pump-in swage from TDS. R/U 10 3/4" spider elevators and function tested same. Installed fill-up and circulation tool.
03:15	03:45	2410	drilling -- casing	ok	P/U 10 3/4" x 9 5/8" x-over joint from HTS using forwards PRS. Unable to lower spider elevator over x-over joint. Fill-up tool sealing element OD to large. L/D fill-up tool. Checked seal element not installed. Decided to continue w/o tool.
03:45	05:00	2410	interruption - repair	ok	Attempted to M/U X-over joint, lower part of X-over joint body 11 3/4" OD. Spun in X-over using chain tongs. Lowered connection in rotary. Made several attempts to M/U connection with Weatherford power tong. Evaluated make-up graph and discussed situation with onshore organization. Accepted connection.
05:00	07:00	2410	drilling -- casing	ok	Removed Weatherford 9 5/8" FMS and installed 10 3/4" PS30 slips. Removed excess equipment from setback area. Changed from spider elevator to BX3 elevator as fill up tool did not fit into casing and spider elevator hanging up on box on 10 3/4" casing. Installed 10 3/4" inserts.
07:00	07:15	2410	drilling -- casing	ok	Performed TBT with new crew for running 10 3/4" casing stands from derrick.
07:15	07:30	2410	drilling -- casing	ok	Function tested handling equipment.
07:30	08:45	2410	drilling -- casing	ok	In preparation for high wind speeds, lifted cement head and HWDP single on HTS and re-arranged equipment on rig floor.
08:45	09:15	2410	interruption - repair	ok	Troubleshoot leaking solenoid valve on TDS.
09:15	10:15	2496	drilling -- casing	ok	RIH with 10 3/4" 65,7 ppf P110 VamTop casing from 2410 m to 2496 m. Filled casing every stand.
10:15	13:45	2902	drilling -- casing	ok	RIH with 10 3/4" casing in from 2496 m to 2902 m. Entered open hole at 2507 m. Filled casing every stand. Average running rate 9,6 joints/hr. Hole in good condition. Took 11 MT weight on last 2 m.
13:45	14:30	2902	drilling -- casing	ok	Changed handling equipment to 5 1/2". P/U and M/U installed casing hanger.
14:30	16:00	2940	drilling -- casing	ok	Removed PS30 slips, installed outer ring. Removed seal assembly protection. Installed master bushing and RIH to 2940 m. Rigged down casing tong.
16:00	16:15	2953	drilling -- casing	ok	P/U 5 1/2" HWDP single from HTS and RIH from 2940 m to 2953 m.
16:15	17:30	3034	drilling -- casing	ok	RIH with 9 5/8" x 10 3/4" casing on 5 1/2" HWDP landing string from 2953 m to 3034 m.
17:30	18:15	3045	drilling -- casing	ok	Took 20 MT weight. Picked up with 35 MT overpull. Established circulation w/ 200 l/min and increased flow to 500 l/min, 20 bar. Washed casing down from 3034 m to 3045 m. Set slip s, B/O TDS, B/O double and racked same.
18:15	19:15	3045	drilling -- casing	ok	P/U cement head from HTS. M/U cement hoses. M/U cement stand to landing string.
19:15	19:30	3047	drilling -- casing	ok	Established circulation with 500 l/min. Landed casing hanger in wellhead as per GE Vetco instructions.
19:30	20:45	3047	drilling -- casing	ok	Increased flow in 250 l/min steps to 1600 l/min/ 37 bar. Held TBT for cement job. Tested rig up prior to cement job.
20:45	21:15	3047	drilling -- casing	ok	Decreased flow to 1500 l/min/ 35 bar due to potential slight losses. Circulated tot. 115 m3.
21:15	21:30	3047	drilling -- casing	ok	Transferred 20 m3 OBM between pits, bypassed sand traps. Displaced cement line with 1 m3 DW.
21:30	21:45	3047	drilling -- casing	ok	Pumped 20 m3 1,60 sg spacer w/ 1500 l/min, 38 bar.
21:45	22:30	3047	drilling -- casing	ok	Confirmed line-up. Dropped bottom dart. Mixed and pumped 15 m3 1,90 sg cement slurry with 630 l/min. Bottom plug released after 900 ltr/119 bar.
22:30	22:45	3047	drilling -- casing	ok	Dropped top dart and displaced same with 1,5 m2 DW from cement unit. Top wiper plug released after 920 ltr/106 bar.
22:45	00:00	3047	drilling -- casing	ok	Displaced cement with 1500 l/min and 14 - 47 bar 1,28 sg OBM using rig pumps. Encountered losses and reduced rate to 1400 l/min, 48 - 54 bar. Reduced rate further to 1300 l/min, 52 - 71 bar. Bottom plug sheared after 4490 strokes with 153 bar. Reduced rate on last 2 m3 to 500 l/min, 53 bar. Bumped top plug after 5170 strokes/112 m3 (94,9 % eff). Lost 8 m3 during displacement.

Equipment Failure Information

Start time	Depth mMD	Depth mTVD	Sub Equip - Syst Class	Operation Downtime (min)	Equipment Repaired	Remark
00:00	3056		service equ -- electr logging eq u	0	00:00	Unable to establish communication between drop gyro tool and computer.

Drilling Fluid

Sample Time	16:00	20:00
Sample Point	Active pit	Flowline
Sample Depth mMD	3056	3056
Fluid Type	Enviromul Yellow	Enviromul Yellow
Fluid Density (g/cm3)	1.28	1.28
Funnel Visc (s)	-999.99	-999.99
Mf ()		
Pm ()		
Pm filtrate ()		
Chloride ()		
Calcium ()		
Magnesium ()		
Ph		
Excess Lime ()		
Solids		
Sand ()		
Water ()		
Oil ()		
Solids ()		
Corrected solids ()		
High gravity solids ()		
Low gravity solids ()		
Viscometer tests		
Plastic visc. (mPa.s)	37	36
Yield point (Pa)	9.5	10
Filtration tests		
Pm filtrate ()		
Filtrate Lthp ()		
Filtrate Hthp ()		
Cake thickn API ()		
Cake thickn HPHT ()		
Test Temp HPHT (degC)	120	120
Comment		

Casing Liner Tubing

Sta rt T ime	00:30	00:30	00:30	00:30	00:30	00:30	00:30	00:30	00:30
End Tim e	16:30	16:30	16:30	16:30	16:30	16:30	16:30	16:30	16:30
Typ e of Pip e	Casing	Casing	Casing	Casing	Casing	Casing	Casing	Casing	Casing
Cas ing Typ e	Production	Production	Production	Production	Production	Production	Production	Production	Production
Out sid e di am eter (in)	10.75	10.75	10.75	9.625	9.625	9.625	9.625	9.625	9.625
Insi de dia met er (i n)	9.56	9.56	9.56	8.535	8.535	8.535	8.535	8.535	8.535
Wei ght (lb m/ft)	65.7	65.7	65.7	53.5	53.5	53.5	53.5	53.5	53.5
Gra de	P-110	P-110	P-110	P-110	P-110	P-110	P-110	P-110	P-110
Con nec tion	Vam TOP	Vam TOP	Vam TOP	Vam TOP	Vam TOP	Vam TOP	Vam TOP	Vam TOP	Vam TOP
Len gth (m)	4.84	479.22	13.22	2371.91	5.9	.66	5.82	12.37	13.03
Top mMD	140.07	144.91	624.1	637.3	3009.21	3015.1	3015.8	3021.6	3033.97

